

BTS SIO – Épreuve E6

Conception et administration d'une infrastructure réseau

Projet Kaizer – Cisco Networklab Infrastructure réseau sécurisée multi-sites

Routage • VLAN • ACL • Sécurisation réseau

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Infrastructure Réseau Sécurisée Multi-Sites

1. Contexte

L'entreprise fictive **Kaizer** est une PME répartie sur deux sites :

- **Site A : Siège**
- **Site B : Agence**

L'entreprise souhaite moderniser son infrastructure réseau afin d'améliorer la communication entre ses différents services, renforcer la sécurité et faciliter l'administration du réseau.

Le projet consiste à concevoir et simuler une **infrastructure réseau sécurisée multi-sites** à l'aide de **Cisco Packet Tracer**.

L'architecture doit être organisée de manière modulaire et évolutive afin de permettre l'ajout futur de nouveaux utilisateurs, équipements ou services.

2. Besoins fonctionnels et techniques

Fonctionnels :

L'infrastructure doit permettre :

- La communication entre les différents services de l'entreprise ;
- La communication entre le Site A et le Site B ;
- La séparation des différents services grâce aux VLAN ;
- La gestion des utilisateurs et des équipements du réseau ;
- Le contrôle des communications entre les différents réseaux ;
- La sécurisation des échanges réseau ;
- La possibilité d'ajouter facilement de nouveaux équipements.

Techniques :

L'infrastructure doit intégrer :

- Des **routeurs Cisco** ;
- Des **commutateurs Cisco** ;
- Plusieurs **VLAN** ;
- Du **routage inter-VLAN** ;
- Une liaison entre le Site A et le Site B ;
- Des mécanismes de filtrage avec des **ACL** ;
- Un plan d'adressage IP organisé ;
- Une architecture évolutive et facilement administrable.

L'ensemble de l'infrastructure est simulé avec **Cisco Packet Tracer**..

3. Réalisation

a. Mise en place de la topologie réseau

La première étape consiste à créer la topologie réseau dans Cisco Packet Tracer.

L'architecture est divisée en deux sites.

Site A – Siège

Le Site A constitue le siège principal de l'entreprise.

Il comprend :

- Un routeur ;
- Un ou plusieurs switches ;
- Des postes utilisateurs ;
- Plusieurs VLAN correspondant aux différents services.

Site B – Agence

Le Site B représente l'agence de l'entreprise.

Il comprend :

- Un routeur ;
- Un switch ;
- Des postes utilisateurs ;
- Des VLAN permettant de séparer les différents services.

Les deux sites sont reliés par une liaison entre les routeurs.

b. Configuration des équipements Cisco

Les équipements Cisco sont configurés directement dans Packet Tracer.

Les routeurs permettent notamment :

- Le routage entre les différents réseaux ;
- La communication entre les deux sites ;
- Le routage inter-VLAN.

Les switches permettent :

- La connexion des postes ;
- La création des VLAN ;
- L'affectation des ports aux différents VLAN ;
- La transmission des VLAN entre les équipements grâce aux liens appropriés.

c. Création des VLAN

VLAN	Service	Utilisation
VLAN10	Administration	Personnel administratif
VLAN20	Utilisateurs	Postes des utilisateurs
VLAN30	Serveurs	Services et serveurs
VLAN40	Management	Administration des équipements

d. Routage inter-VLAN

Le routage inter-VLAN permet aux différents réseaux de communiquer lorsque cela est nécessaire.

Le routeur est configuré afin de gérer les communications entre les VLAN.

La configuration permet notamment :

- Aux utilisateurs autorisés d'accéder aux services nécessaires ;
- De contrôler les communications entre les différents VLAN ;
- De maintenir une séparation logique entre les services.

Des tests de ping sont réalisés afin de vérifier le fonctionnement du routage.

e. Liaison entre le Site A et le Site B

Une liaison réseau est mise en place entre les deux routeurs.

Elle permet aux équipements du Site A de communiquer avec ceux du Site B.

L'adressage IP est configuré sur les interfaces des routeurs.

Le routage est ensuite configuré afin que les différents réseaux puissent être atteints depuis les deux sites.

Cette architecture permet de simuler le fonctionnement d'une entreprise possédant plusieurs implantations géographiques.

f. Sécurisation du réseau

La sécurité du réseau est renforcée grâce à plusieurs mécanismes.

VLAN

Les VLAN permettent d'isoler logiquement les différents services.

Par exemple, les postes administratifs ne se trouvent pas dans le même réseau logique que les postes utilisateurs.

ACL

Des **Access Control Lists (ACL)** sont configurées sur les routeurs afin de contrôler les communications.

Elles permettent notamment de :

- Autoriser certains flux ;
- Bloquer certains flux ;
- Limiter les communications entre certains VLAN ;
- Contrôler les échanges entre les deux sites.

Sécurisation des équipements

Les équipements Cisco peuvent également être protégés par :

- Des mots de passe ;
- La sécurisation des accès administrateur ;
- La désactivation des ports inutilisés ;
- La configuration des ports nécessaires uniquement.

4. Résultats & tests

Plusieurs tests sont réalisés dans Cisco Packet Tracer afin de vérifier le fonctionnement de l'infrastructure.

Test 1 – Communication dans un même VLAN

Un poste utilisateur communique avec un autre poste appartenant au même VLAN.

Résultat : communication fonctionnelle.

Test 2 – Routage inter-VLAN

Un poste d'un VLAN autorisé communique avec un autre VLAN.

Résultat : communication fonctionnelle après configuration du routage inter-VLAN.

Test 3 – Communication entre les deux sites

Un poste du Site A communique avec un poste du Site B.

Résultat : communication fonctionnelle après configuration du routage entre les sites.

Test 4 – Filtrage ACL

Un flux interdit par une ACL est testé.

Résultat : le trafic non autorisé est bloqué.

Test 5 – Vérification de la configuration

Les commandes Cisco permettent de vérifier :

- Les VLAN ;
- Les interfaces ;
- Les routes ;
- Les ACL ;
- L'état des équipements.

5. Analyse personnelle

Ce projet m'a permis de développer mes compétences en administration réseau et en configuration des équipements Cisco.

J'ai notamment appris à :

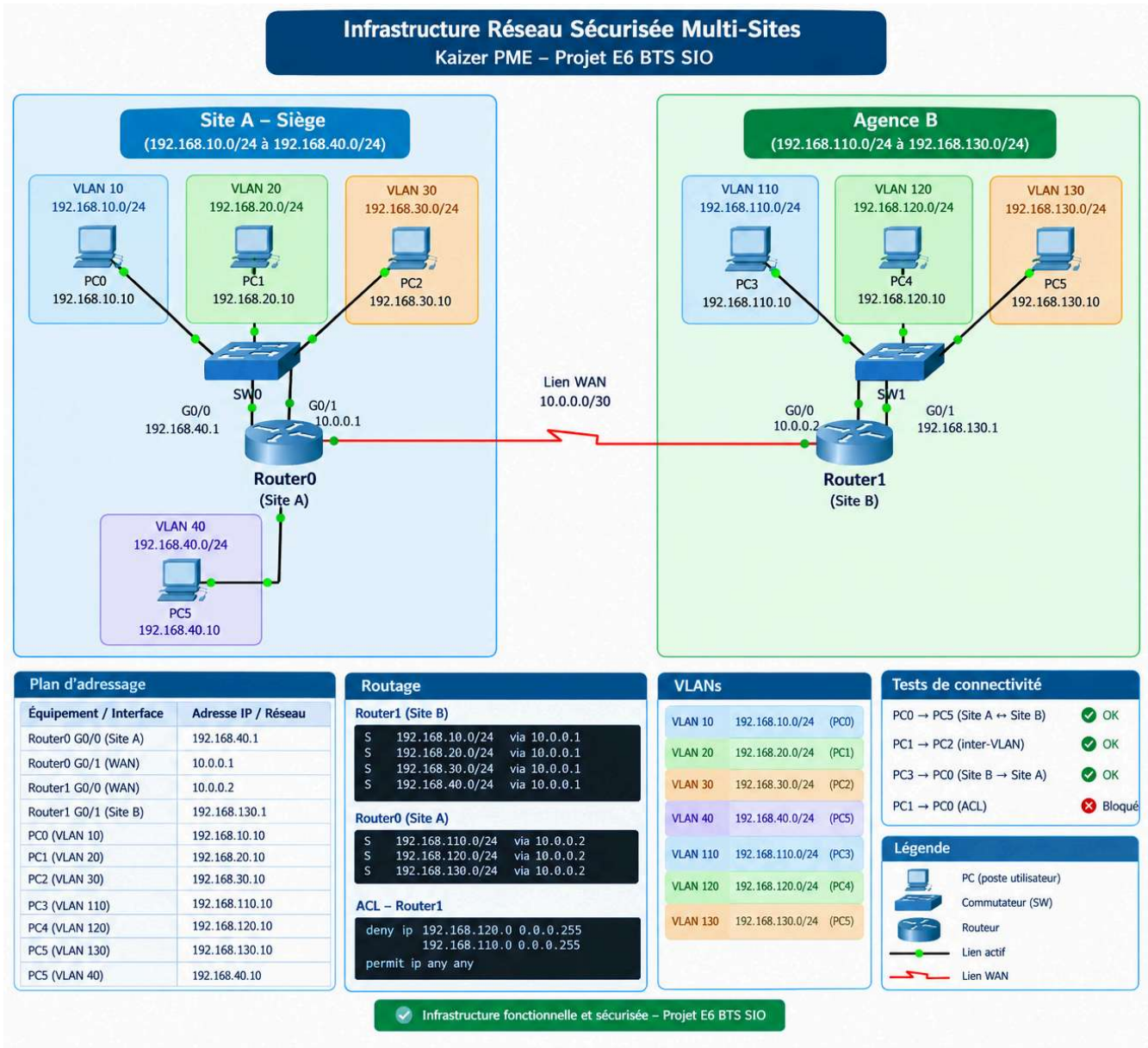
- Concevoir une architecture réseau multi-sites ;
- Utiliser Cisco Packet Tracer ;
- Configurer des switches et des routeurs ;
- Créer et administrer des VLAN ;
- Configurer le routage inter-VLAN ;
- Mettre en place une communication entre deux sites ;
- Configurer des règles ACL ;
- Tester la connectivité avec la commande **ping** ;
- Vérifier une configuration réseau avec les commandes Cisco ;
- Identifier et corriger des problèmes de communication réseau.

Les principales difficultés ont concerné la configuration du routage, des VLAN et des règles de filtrage.

Certains problèmes de communication peuvent provenir d'une mauvaise configuration d'adresse IP, d'un port mal affecté à un VLAN, d'une interface non activée ou d'une route manquante.

La résolution de ces problèmes nécessite une vérification progressive de la configuration des switches, des routeurs et des tables de routage.

6. Schéma réseau



7. Liste de screenshots

Topologie

- 1. Vue générale de la topologie Packet Tracer.
- 2. Vue détaillée du Site A et B.

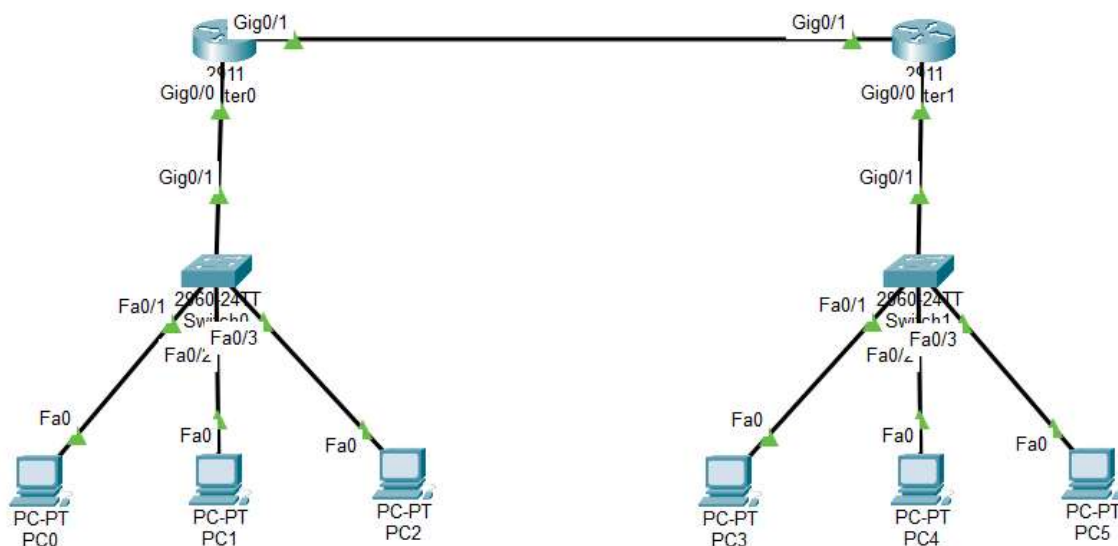
Press RETURN to get started!

```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#
```



VLAN et switches

3. Création des VLAN.


```

Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name ADMIN
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name USERS
Switch(config-vlan)#exit
Switch(config)#vlan 30
Switch(config-vlan)#name SERVICES
Switch(config-vlan)#exit
Switch(config)#vlan 40
Switch(config-vlan)#name MANAGEMENT
Switch(config-vlan)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	ADMIN	active	
20	USERS	active	
30	SERVICES	active	
40	MANAGEMENT	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

4. Affectation des ports aux VLAN.

```

Switch#enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fastEthernet 0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#interface fastEthernet 0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#

```

5. Configuration des trunks.

```

Switch(config)#interface gigabitEthernet 0/1
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Gig0/1    1-1005

Port      Vlans allowed and active in management domain
Gig0/1    1,10,20,30,40

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1    none

```

Routage

6. Configuration des interfaces du routeur du Site A et B.

```

Router#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#
Router(config-if)#exit

```

7. Configuration du routage inter-VLAN.

```

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface gigabitEthernet 0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up
encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.110.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.120.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.30
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
encapsulation dot1Q 30
Router(config-subif)#ip address 192.168.130.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.40
Router(config-subif)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.40, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up
encapsulation dot1Q 40
Router(config-subif)#ip address 192.168.140.1 255.255.255.0
Router(config-subif)#exit
Router(config)#end

show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	unassigned	YES	manual	up	up
GigabitEthernet0/0.10	192.168.110.1	YES	manual	up	up
GigabitEthernet0/0.20	192.168.120.1	YES	manual	up	up
GigabitEthernet0/0.30	192.168.130.1	YES	manual	up	up
GigabitEthernet0/0.40	192.168.140.1	YES	manual	up	up
GigabitEthernet0/1	10.0.0.2	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```

Router#

```

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::290:2BFF:FEED:E920

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

8. Configuration entre les routeurs

```

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.110.0 255.255.255.0 10.0.0.2
Router(config)#ip route 192.168.120.0 255.255.255.0 10.0.0.2
Router(config)#ip route 192.168.130.0 255.255.255.0 10.0.0.2
Router(config)#ip route 192.168.140.0 255.255.255.0 10.0.0.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
    192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
    192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L       192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
S       192.168.110.0/24 [1/0] via 10.0.0.2
S       192.168.120.0/24 [1/0] via 10.0.0.2
S       192.168.130.0/24 [1/0] via 10.0.0.2
S       192.168.140.0/24 [1/0] via 10.0.0.2

```

9. Table de routage avec `show ip route`.

show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, GigabitEthernet0/1
L    10.0.0.1/32 is directly connected, GigabitEthernet0/1
192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L    192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L    192.168.30.1/32 is directly connected, GigabitEthernet0/0.30
192.168.40.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.40.0/24 is directly connected, GigabitEthernet0/0.40
L    192.168.40.1/32 is directly connected, GigabitEthernet0/0.40
S    192.168.110.0/24 [1/0] via 10.0.0.2
S    192.168.120.0/24 [1/0] via 10.0.0.2
S    192.168.130.0/24 [1/0] via 10.0.0.2
S    192.168.140.0/24 [1/0] via 10.0.0.2
```

show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    10.0.0.0/8 is directly connected, GigabitEthernet0/1
L    10.0.0.2/32 is directly connected, GigabitEthernet0/1
S    192.168.10.0/24 [1/0] via 10.0.0.1
S    192.168.20.0/24 [1/0] via 10.0.0.1
S    192.168.30.0/24 [1/0] via 10.0.0.1
S    192.168.40.0/24 [1/0] via 10.0.0.1
192.168.110.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.110.0/24 is directly connected, GigabitEthernet0/0.10
L    192.168.110.1/32 is directly connected, GigabitEthernet0/0.10
192.168.120.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.120.0/24 is directly connected, GigabitEthernet0/0.20
L    192.168.120.1/32 is directly connected, GigabitEthernet0/0.20
192.168.130.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.130.0/24 is directly connected, GigabitEthernet0/0.30
L    192.168.130.1/32 is directly connected, GigabitEthernet0/0.30
192.168.140.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.140.0/24 is directly connected, GigabitEthernet0/0.40
L    192.168.140.1/32 is directly connected, GigabitEthernet0/0.40
```

Communication entre les sites.

10. Test **ping** entre Site A et Site B.

```
C:\>ping 192.168.110.10

Pinging 192.168.110.10 with 32 bytes of data:

Reply from 192.168.110.10: bytes=32 time<1ms TTL=126
Reply from 192.168.110.10: bytes=32 time=1ms TTL=126
Reply from 192.168.110.10: bytes=32 time<1ms TTL=126
Reply from 192.168.110.10: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.110.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

11. Test de communication entre différents VLAN.

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=70ms TTL=127
Reply from 192.168.20.10: bytes=32 time=17ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 70ms, Average = 21ms

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time=1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127
Reply from 192.168.10.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

```

C:\>ping 192.168.130.10

Pinging 192.168.130.10 with 32 bytes of data:

Reply from 192.168.130.10: bytes=32 time<1ms TTL=127
Reply from 192.168.130.10: bytes=32 time<1ms TTL=127
Reply from 192.168.130.10: bytes=32 time<1ms TTL=127
Reply from 192.168.130.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.130.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.120.10

Pinging 192.168.120.10 with 32 bytes of data:

Reply from 192.168.120.10: bytes=32 time<1ms TTL=127
Reply from 192.168.120.10: bytes=32 time<1ms TTL=127
Reply from 192.168.120.10: bytes=32 time<1ms TTL=127
Reply from 192.168.120.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.120.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.110.10

Pinging 192.168.110.10 with 32 bytes of data:

Reply from 192.168.110.10: bytes=32 time<1ms TTL=127
Reply from 192.168.110.10: bytes=32 time<1ms TTL=127
Reply from 192.168.110.10: bytes=32 time<1ms TTL=127
Reply from 192.168.110.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.110.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.130.10

Pinging 192.168.130.10 with 32 bytes of data:

Reply from 192.168.130.10: bytes=32 time<1ms TTL=127
Reply from 192.168.130.10: bytes=32 time<1ms TTL=127
Reply from 192.168.130.10: bytes=32 time=1ms TTL=127
Reply from 192.168.130.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.130.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>

```

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.110.10

Pinging 192.168.110.10 with 32 bytes of data:

Reply from 192.168.110.10: bytes=32 time<1ms TTL=127
Reply from 192.168.110.10: bytes=32 time<1ms TTL=127
Reply from 192.168.110.10: bytes=32 time<1ms TTL=127
Reply from 192.168.110.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.110.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.120.10

Pinging 192.168.120.10 with 32 bytes of data:

Reply from 192.168.120.10: bytes=32 time<1ms TTL=127
Reply from 192.168.120.10: bytes=32 time<1ms TTL=127
Reply from 192.168.120.10: bytes=32 time<1ms TTL=127
Reply from 192.168.120.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.120.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|

```

Sécurité

12. Configuration des ACL.

```

Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#access-list 100 deny ip 192.168.120.0 0.0.0.255 192.168.110.0 0.0.0.255
Router(config)#access-list 100 permit ip any any
Router(config)#interface gigabitEthernet 0/0.20
Router(config-subif)#ip access-group 100 in
Router(config-subif)#exit
Router(config)#end
Router#
%SYS-5-CONFIG I: Configured from console by console
show ip interface gigabitEthernet 0/0.20
GigabitEthernet0/0.20 is up, line protocol is up (connected)
  Internet address is 192.168.120.1/24
  Broadcast address is 255.255.255.255
  Address determined by setup command
  MTU is 1500 bytes
  Helper address is not set
  Directed broadcast forwarding is disabled
  Outgoing access list is not set
  Inbound access list is 100
  Proxy ARP is enabled
  Security level is default
  Split horizon is enabled
  ICMP redirects are always sent
  ICMP unreachables are always sent
  ICMP mask replies are never sent
  IP fast switching is disabled
  IP fast switching on the same interface is disabled
  IP Flow switching is disabled
  IP Fast switching turbo vector
  IP multicast fast switching is disabled
  IP multicast distributed fast switching is disabled
  Router Discovery is disabled

```

13. Vérification des ACL avec `show access-lists`.

```
Router#show access-lists
Extended IP access list 100
 10 deny ip 192.168.120.0 0.0.0.255 192.168.110.0 0.0.0.255
 20 permit ip any any
```

14. Test d'un trafic autorisé et bloqué.

```
C:\>ping 192.168.110.10

Pinging 192.168.110.10 with 32 bytes of data:

Reply from 192.168.120.1: Destination host unreachable.
Reply from 192.168.120.1: Destination host unreachable.
Reply from 192.168.120.1: Destination host unreachable.
Reply from 192.168.120.1: Destination host unreachable.

Ping statistics for 192.168.110.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.130.10

Pinging 192.168.130.10 with 32 bytes of data:

Reply from 192.168.130.10: bytes=32 time<1ms TTL=127
Reply from 192.168.130.10: bytes=32 time=1ms TTL=127
Reply from 192.168.130.10: bytes=32 time<1ms TTL=127
Reply from 192.168.130.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.130.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:

Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.
Reply from 192.168.20.1: Destination host unreachable.

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:

Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Validation

16. Vérification finale de la configuration des switches.

Switch#show vlan brief

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10	ADMIN	active	Fa0/1
20	USERS	active	Fa0/2
30	SERVICES	active	Fa0/3
40	MANAGEMENT	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Switch#show interfaces trunk

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Gig0/1	1-1005

Port	Vlans allowed and active in management domain
Gig0/1	1,10,20,30,40

Port	Vlans in spanning tree forwarding state and not pruned
Gig0/1	1,10,20,30,40

Switch#show running-config

Building configuration...

Current configuration : 1256 bytes

```
!  
version 15.0  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
no service password-encryption  
!  
hostname Switch  
!  
!  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface FastEthernet0/1  
  switchport access vlan 10  
  switchport mode access
```

```

Switch#show vlan brief

```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/2
10	ADMIN	active	Fa0/1
20	USERS	active	Fa0/2
30	SERVICES	active	Fa0/3
40	MANAGEMENT	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```

Switch#show interfaces trunk

```

Port	Mode	Encapsulation	Status	Native vlan
Gig0/1	on	802.1q	trunking	1

```

Switch#show interfaces trunk

```

Port	Vlans allowed on trunk
Gig0/1	1-1005

```

Switch#show interfaces trunk

```

Port	Vlans allowed and active in management domain
Gig0/1	1,10,20,30,40

```

Switch#show interfaces trunk

```

Port	Vlans in spanning tree forwarding state and not pruned
Gig0/1	1,10,20,30,40

```

Switch#show running-config
Building configuration...

Current configuration : 1256 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Switch
!
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
switchport access vlan 10
switchport mode access

```

17. Vérification finale de la configuration des routeurs.

```

Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       unassigned      YES manual up          up
GigabitEthernet0/0.10    192.168.10.1    YES manual up          up
GigabitEthernet0/0.20    192.168.20.1    YES manual up          up
GigabitEthernet0/0.30    192.168.30.1    YES manual up          up
GigabitEthernet0/0.40    192.168.40.1    YES manual up          up
GigabitEthernet0/1       10.0.0.1        YES manual up          up
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
    192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.10.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.10.1/32 is directly connected, GigabitEthernet0/0.10
    192.168.20.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.20.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.20.1/32 is directly connected, GigabitEthernet0/0.20
    192.168.30.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.30.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.30.1/32 is directly connected, GigabitEthernet0/0.30

Router#show access-lists
Extended IP access list 100
    10 deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255 (4 match(es))
    20 permit ip any any (4 match(es))

Router#show running-config
Building configuration...

Current configuration : 1393 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!
!

```

```

Router#show ip interface brief
Interface          IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0 unassigned      YES manual up          up
GigabitEthernet0/0.10 192.168.110.1 YES manual up          up
GigabitEthernet0/0.20 192.168.120.1 YES manual up          up
GigabitEthernet0/0.30 192.168.130.1 YES manual up          up
GigabitEthernet0/0.40 192.168.140.1 YES manual up          up
GigabitEthernet0/1    10.0.0.2        YES manual up          up
GigabitEthernet0/2    unassigned      YES unset  administratively down down
Vlan1                 unassigned      YES unset  administratively down down

Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, GigabitEthernet0/1
L       10.0.0.2/32 is directly connected, GigabitEthernet0/1
S       192.168.10.0/24 [1/0] via 10.0.0.1
S       192.168.20.0/24 [1/0] via 10.0.0.1
S       192.168.30.0/24 [1/0] via 10.0.0.1
S       192.168.40.0/24 [1/0] via 10.0.0.1
    192.168.110.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.110.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.110.1/32 is directly connected, GigabitEthernet0/0.10
    192.168.120.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.120.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.120.1/32 is directly connected, GigabitEthernet0/0.20
    192.168.130.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.130.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.130.1/32 is directly connected, GigabitEthernet0/0.30
    192.168.140.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.140.0/24 is directly connected, GigabitEthernet0/0.40
L       192.168.140.1/32 is directly connected, GigabitEthernet0/0.40

Router#show access-lists
Extended IP access list 100
    10 deny ip 192.168.120.0 0.0.0.255 192.168.110.0 0.0.0.255 (4 match(es))
    20 permit ip any any (4 match(es))

Router#show access-lists
Extended IP access list 100
    10 deny ip 192.168.120.0 0.0.0.255 192.168.110.0 0.0.0.255 (4 match(es))
    20 permit ip any any (4 match(es))

Router#show running-config
Building configuration...

Current configuration : 1395 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
!
!
!
ip cef
no ipv6 cef
!
!

```

18. Vue finale de l'infrastructure fonctionnelle

